

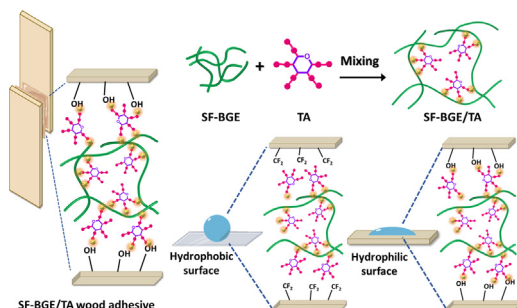
Eco-friendly silk fibroin/tannic acid coacervates for humid and underwater wood adhesives

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GRAPHICAL ABSTRACT



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ABSTRACT

Bioadhesives derived from biomass are steadily gaining spotlight as substitutes for formaldehyde-based resins in the adhesive industry. However, there is a need to develop novel water-resistant bioadhesives with high adhesive and cohesive strengths because the currently available biomaterial-based adhesives have low mechanical strength. In this study, a complex coacervate was prepared easily by mixing silk fibroin and tannic acid to produce a bioadhesive with high adhesive and cohesive strengths as well as water resistance. The silk fibroin–tannic acid coacervate adhered well to various substrates, and its adhesive strength according to the type of substrate and water contact angle were evaluated comparatively. In particular, the adhesive strength of this adhesive on a wood substrate was systematically analyzed by varying different experimental parameters (relative humidity, surface roughness of the substrate, water stability, and pH). This cost-effective coacervate is applicable as an eco-friendly wood adhesive.

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1. Introduction

The wood-based panel industry largely uses thermosetting adhesives derived from synthetic petroleum chemicals, including formaldehyde, polyurethane, epoxies, urea, melamine, and phenol. However, these adhesives cause environmental pollution because

they emit carcinogenic formaldehyde and other volatile organic compounds [1–4]. Owing to global government regulations and changes in the public perception of environmental pollution, sustainable and eco-friendly polymer-based bioadhesives are being developed to overcome the shortcomings of wood adhesives, and to reduce or eliminate environmental pollution [5,6].

Many studies have aimed to produce eco-friendly wood adhesives using bio-based materials (lignin, tannin, protein, and polysaccharides), and have achieved good adhesion performances [7–11]. For example, Guo *et al.* reported an inexpensive and eco-

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